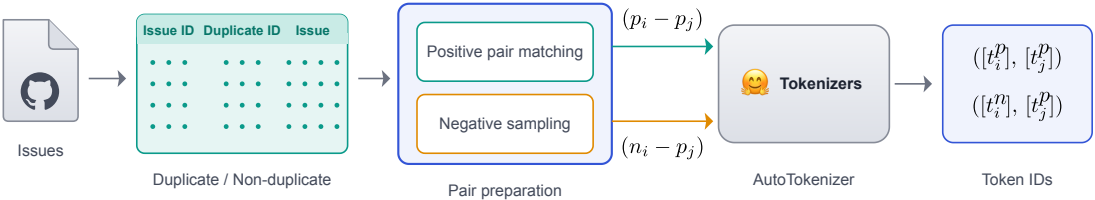
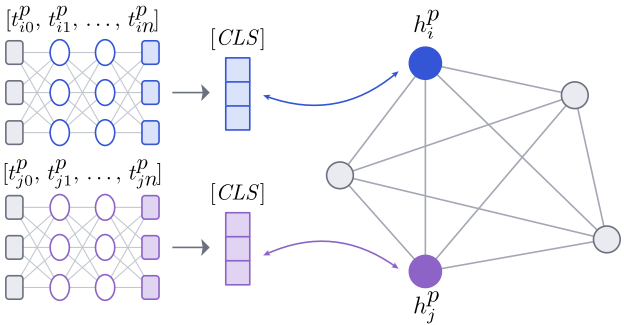
Data preprocessing



Embedding construction and training



Positive pair training

$$\hat{\boldsymbol{y}} = \mathrm{sim}(h_i^p, h_j^p)$$

$$\mathcal{L}_{\mathrm{BCE}}(1, \hat{\boldsymbol{y}}) = -[\log(\hat{\boldsymbol{y}})]$$

Negative pair training

$$\hat{\boldsymbol{y}} = \mathrm{sim}(h_i^n, h_j^p)$$

$$\mathcal{L}_{\mathrm{BCE}}(0, \hat{\boldsymbol{y}}) = -[\log(1 - \hat{\boldsymbol{y}})]$$